3.1.7 Digital design and manufacture

Computer aided design (CAD)

Content	Potential links to maths and science
• The advantages and disadvantages of using CAD compared to manually generated alternative. • The use of CAD to develop and present ideas for products. • How CAD is used in industrial applications.	Use of datum points and geometry when setting out design drawings. The use of tolerances in dimensioning.

Computer aided manufacturing (CAM)

Content	Potential links to maths and science
Students should be aware of, and be able to describe, how CAM is used in the manufacture of products. Specific processes to include: • fabric manufacture • fabric printing • lay planning and computer controlled cutting • automated buttonholing • making and sewing of pockets • seam stitching • pressing • computer controlled decorative processes • laser cutting.	Calculating speeds and times for machining.

Virtual modelling

Content	Potential links to maths and science
Students should be aware of, and be able to describe, how virtual modelling/testing is used in industry prior to product production. Specific processes to include: • simulation • pattern design systems • computer controlled printing to produce sample fabric lengths.	

Electronic data interchange

Content	Potential links to maths and science
Students should be aware of, and able to describe, the use of electronic point of sales (EPOS) for marketing purposes and the collection of market research data.	

Production, planning and control (PPC) networking

Content	Potential links to maths and science
Students should be aware of, and able to describe, the role of PPC systems in the planning and controlling of all aspects of manufacturing, including: • availability of materials • scheduling of machines and people • coordinating suppliers and customers.	